Problem Set 2

Applied Stats/Quant Methods 1

Due: October 23, 2025

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in R, please include the code you used to get your answers. Please also include the .R file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday October 23, 2025. No late assignments will be accepted.

Question 1: Political Science

The following table was created using the data from a study run in a major Latin American city.[1] As part of the experimental treatment in the study, one employee of the research team was chosen to make illegal left turns across traffic to draw the attention of the police officers on shift. Two employee drivers were upper class, two were lower class drivers, and the identity of the driver was randomly assigned per encounter. The researchers were interested in whether officers were more or less likely to solicit a bribe from drivers depending on their class (officers use phrases like, "We can solve this the easy way" to draw a bribe). The table below shows the resulting data.

[1] Fried, Lagunes, and Venkataramani (2010). "Corruption and Inequality at the Crossroad: A Multi-method Study of Bribery and Discrimination in Latin America. *Latin American Research Review*. 45 (1): 76-97.

	Not Stopped	Bribe requested	Stopped/given warning
Upper class	14	6	7
Lower class	7	7	1

(a) Calculate the χ^2 test statistic by hand/manually (even better if you can do "by hand" in `R`).

Step 1: Put the data in R and calculate the totals

First I put the tabular data into a matrix. The table without totals or labels was named observed_table. Then I added the row and column totals and labeled the matrix the same way the original table was labeled. I named the table with the totals and labels class_table

```r
observed_table <- matrix(
  c(14, 6, 7, 7, 7, 1),
  nrow = 2,
  ncol = 3,
  byrow = TRUE
)

table_row_totals <- rbind(observed_table,
                          c(sum(observed_table[ ,1]),
                            sum(observed_table[ ,2]),
                            sum(observed_table[ ,3])))

class_table <- cbind(table_row_totals,
                     c(sum(table_row_totals[1, ]),
                       sum(table_row_totals[2, ]),
                       sum(table_row_totals[3, ])))


rownames(class_table) <- c("Upper Class", "Lower Class", "Column Total")
colnames(class_table) <- c("Not Stopped", "Bribe Requested",
                           "Stoped/Given Warning", "Row Total")
class_table
```

```
             Not Stopped Bribe Requested Stoped/Given Warning Row Total
Upper Class           14               6                    7        27
Lower Class            7               7                    1        15
Column Total          21              13                    8        42
```

Step 2: Calculate the expected values

Find the expected values by calculating row total/grand total * column total

```
expected_values <- matrix(c(
  (class_table[1,4] / class_table[3,4] * class_table[3,1]),
  (class_table[1,4] / class_table[3,4] * class_table[3,2]),
  (class_table[1,4] / class_table[3,4] * class_table[3,3]),
  (class_table[2,4] / class_table[3,4] * class_table[3,1]),
  (class_table[2,4] / class_table[3,4] * class_table[3,2]),
  (class_table[2,4] / class_table[3,4] * class_table[3,3])),
   nrow = 2,
   ncol = 3,
   byrow = TRUE
  )

rownames(expected_values) <- c("Upper Class", "Lower Class")
colnames(expected_values) <- c("Not Stopped", "Bribe Requested",
                              "Stopped/Given Warning")
expected_values
```

```
            Not Stopped Bribe Requested Stopped/Given Warning
Upper Class        13.5        8.357143              5.142857
Lower Class         7.5        4.642857              2.857143
```

Step 3: Assumptions

- The data is categorical
- The observations are independent
- The groups are mutually exclusive
- All of the expected counts are at least 5 (that isn't met in this example)

Step 4: State hypotheses

- Null Hypothesis: The variables are statistically independent
- Alternative Hypothesis: The variables are statistically dependent

Step 5: Calculate the Chi Squared Statistic

Calculate the Chi-Squared Statistic by taking the (observed values minus the expected values) squared, then dividing that by the expected value. Do that for every value in the table then sum all of them together

```
test_statistic <- sum(((observed_table - expected_values)^2)
                     / expected_values)
test_statistic
```

```
[1] 3.791168
```

(b) Now calculate the p-value from the test statistic you just created (in R).[2] What do you conclude if $\alpha = 0.1$?

Step 6: Calculate the degrees of freedom

Find the degrees of freedom by doing (Rows - 1) * (Columns - 1)

```
class_df <- (nrow(observed_table) - 1) * (ncol(observed_table) - 1)
class_df
```

```
[1] 2
```

Step 7: Calculate the p-value

Input the test statistic and degrees of freedom in R to get the p-value

```
pchisq(test_statistic, df=class_df, lower.tail=FALSE)
```

```
[1] 0.1502306
```

The p-value of 0.1502306 is greater than the critical value of 0.1. Therefore, we fail to reject the null hypothesis that the variables are statistically independent.

[2]Remember frequency should be > 5 for all cells, but let's calculate the p-value here anyway.

(c) Calculate the standardized residuals for each cell and put them in the table below.

Step 8: Calculate the standardized residuals

Calculate the standardized residuals by doing the observed minus the expected values divided by the square root of the (expected frequency * (1 - the row proportion) * (1 - the column proportion))

Cell [1,1]

```
difference <- observed_table - expected_values
difference[1,1] / sqrt(expected_values[1,1]*
          (1 - class_table[1,4] / class_table[3,4])*
          (1 - class_table[3,1]/class_table[3,4]))
```

```
[1] 0.3220306
```

Cell [1,2]

```
difference[1,2] / sqrt(expected_values[1,2]*
          (1 - class_table[1,4] / class_table[3,4])*
          (1 - class_table[3,2]/class_table[3,4]))
```

```
[1] -1.641957
```

Cell [1,3]

```
difference[1,3] / sqrt(expected_values[1,3]*
          (1 - class_table[1,4] / class_table[3,4])*
          (1 - class_table[3,3]/class_table[3,4]))
```

```
[1] 1.523026
```

Cell [2,1]

```
difference[2,1] / sqrt(expected_values[2,1]*
          (1 - class_table[2,4] / class_table[3,4])*
          (1 - class_table[3,1]/class_table[3,4]))
```

```
[1] -0.3220306
```

Cell [2,2]

```
difference[2,2] / sqrt(expected_values[2,2]*
          (1 - class_table[2,4] / class_table[3,4])*
          (1 - class_table[3,2]/class_table[3,4]))
```

```
[1] 1.641957
```

Cell [2,3]

```
difference[2,3] / sqrt(expected_values[2,3]*
          (1 - class_table[2,4] / class_table[3,4])*
          (1 - class_table[3,3]/class_table[3,4]))
```

```
[1] -1.523026
```

	Not Stopped	Bribe requested	Stopped/given warning
Upper class	0.3220306	-1.641957	1.523026
Lower class	-0.3220306	1.641957	-1.523026

(d) How might the standardized residuals help you interpret the results?

Since none of our standard residuals have an absolute value over 2, we can determine that there are no values significantly outside of what we would expect based on our null hypothesis that the variables are statistically independent. This agrees with the information from our p-value that we cannot reject the null hypothesis that the variables are statistically independent.

Question 2: Economics

Chattopadhyay and Duflo were interested in whether women promote different policies than men.[3] Answering this question with observational data is pretty difficult due to potential confounding problems (e.g. the districts that choose female politicians are likely to systematically differ in other aspects too). Hence, they exploit a randomized policy experiment in India, where since the mid-1990s, $\frac{1}{3}$ of village council heads have been randomly reserved for women. A subset of the data from West Bengal can be found at the following link: `https://raw.githubusercontent.com/kosukeimai/qss/master/PREDICTION/women.csv`

Each observation in the data set represents a village and there are two villages associated with one GP (i.e. a level of government is called "GP"). Figure 1 below shows the names and descriptions of the variables in the dataset. The authors hypothesize that female politicians are more likely to support policies female voters want. Researchers found that more women complain about the quality of drinking water than men. You need to estimate the effect of the reservation policy on the number of new or repaired drinking water facilities in the villages.

Figure 1: Names and description of variables from Chattopadhyay and Duflo (2004).

Name	Description
`GP`	An identifier for the Gram Panchayat (GP)
`village`	identifier for each village
`reserved`	binary variable indicating whether the GP was reserved for women leaders or not
`female`	binary variable indicating whether the GP had a female leader or not
`irrigation`	variable measuring the number of new or repaired irrigation facilities in the village since the reserve policy started
`water`	variable measuring the number of new or repaired drinking-water facilities in the village since the reserve policy started

[3] Chattopadhyay and Duflo. (2004). "Women as Policy Makers: Evidence from a Randomized Policy Experiment in India. *Econometrica.* 72 (5), 1409-1443.

(a) State a null and alternative (two-tailed) hypothesis.

Step 1: State Hypotheses

1. Null hypothesis: the reservation policy has no correlation with the number of new or repaired drinking-water facilities

2. Alternative hypothesis: the reservation policy has a correlation with the number of new or repaired drinking-water facilities

(b) Run a bivariate regression to test this hypothesis in R (include your code!).

Step 2: Run a bivariate regression

Use the lm function in R to run the regression, I subset two columns from the original data set to input only the two variables that I am investigating the relationship between, reserved and water. With reserved as the explanatory variable and water as the response variable.

```r
women <- read.csv("women.csv")
bivariate_regression <- lm(women$water ~ women$reserved)
bivariate_regression
```

```
Call:
lm(formula = women$water ~ women$reserved)

Coefficients:
(Intercept)  women$reserved
14.738           9.252
```

Summary of the regression output

```r
summary(bivariate_regression)
```

```

Call:
lm(formula = women$water ~ women$reserved)

Residuals:
Min      1Q  Median      3Q     Max
-23.991 -14.738  -7.865   2.262 316.009

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)      14.738      2.286   6.446 4.22e-10 ***
women$reserved    9.252      3.948   2.344   0.0197 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 33.45 on 320 degrees of freedom
Multiple R-squared:  0.01688, Adjusted R-squared:  0.0138
F-statistic: 5.493 on 1 and 320 DF,  p-value: 0.0197
```

(c) Interpret the coefficient estimate for reservation policy.

Interpret the coefficient estimate

The coefficient estimate for reservation policy of 9.252 is positive, suggesting a positive linear association between reservation policy and new or repaired water facilities. The coefficient estimate of 9.252 indicates that for a one unit increase in reservation policy (in this case a one unit increase represents going from no reservation policy to having a reservation policy) there is a 9.252 increase in the number of new or repaired drinking water facilities in the village